

Stakeholder 18

Cleaning and Caretaking Teams

80 practical questions for privacy-preserving, evidence-disciplined building support

80

practical questions

8

operational categories

6

substantive fields per question

PROPOSED

sensing until independently verified

Scope and operating principles

Purpose

Practical cleaning and caretaking questions linked to proposed evidence, current authoritative sources, explicit analysis, strict answer boundaries and concrete operational actions.

Proposed evidence, not assumed infrastructure

Every sensor, integration and analytic remains proposed until deployment, asset/location mapping, commissioning, permission and current health are independently verified. Missing, unmapped, stale or unhealthy evidence is unknown - not zero.

Historical plan boundary

The May 2021 floor plan is historical spatial reference only. It cannot establish current room use, access, services, hazards, closures, stores, routes, waste points or cleaning responsibility. Current authoritative owners take precedence.

People and privacy boundary

Do not identify or track people, build movement histories, infer behaviour, productivity, health, disability or other protected traits, monitor conversations, or inspect research, screens or network content. Use only the minimum room/zone/asset precision needed.

Safety and authority boundary

Do not grant access, issue permits, prescribe chemicals, dilution or PPE, classify unknown hazards, certify cleanliness or safety, operate or maintain equipment, reopen spaces, close another owner's incident, or replace current cleaning standards, COSHH/SDS, risk assessments, security rules, emergency procedures, competent professionals or supervisor judgement.

Evidence discipline and honest failure

Telemetry answers state modality/location, coverage, sampling/aggregation, validation/calibration, clock alignment, uptime, gaps, drift/outliers and provenance. Owner records state precedence, effective/observed/retrieved times, permissions, provenance, staleness and conflicts. When decisive evidence is missing, the right answer can be "cannot determine" plus the authorised verification route.

DECISION PRINCIPLE

Formal release, active restrictions, controlled documents, competent direct observations within scope and responsible human judgement outrank telemetry. Telemetry can prioritise timing and attention; it cannot create permission, authority, diagnosis or certification.

Field guide

1

Stakeholder question

A natural role-specific decision about a shift, route, room release, washroom, incident, access, equipment, stock, waste, quality or handover.

2

Sensors and telemetry

Proposed physically plausible modality/location, coverage, sampling/aggregation, calibration or functional checks, clock alignment, uptime, gaps, drift and provenance.

3

Authoritative supporting data

Current owners and precedence, effective/observed/retrieved times, permissions, provenance, staleness and conflict handling. May 2021 plan remains historical only.

4

Analysis required

Explicit operation type - lookup, observation, calculation, estimate, forecast, comparison, diagnosis or recommendation - plus interval, minimum evidence, method, uncertainty and safe failure.

5

Answer boundary

What the service must not infer, disclose, promise, certify, authorise, prescribe, operate or control.

6

Why this is useful

The concrete sequencing, service, escalation, replenishment, route, handover or planning action enabled by the answer.

READING RULE

Keep current authoritative record, direct operational observation, proposed telemetry, derived analysis and historical spatial reference visibly distinct. A plausible estimate never becomes permission or authority because it has a confidence score.

Catalogue map

1. SHIFTS, ZONES, STANDARDS AND DAILY PRIORITIES

CT-001 to CT-010 | 10 questions

2. OCCUPANCY-DRIVEN DEMAND AND ROOM-RELEASE INFORMATION

CT-011 to CT-020 | 10 questions

3. WASHROOMS, CONSUMABLES AND AMENITIES

CT-021 to CT-030 | 10 questions

4. SPILLS, DAMAGE, FAULTS AND URGENT ESCALATION

CT-031 to CT-040 | 10 questions

5. KEYS, ACCESS, LONE WORKING AND SECURITY BOUNDARIES

CT-041 to CT-050 | 10 questions

6. EQUIPMENT, CHEMICALS, STORAGE AND REPLENISHMENT

CT-051 to CT-060 | 10 questions

7. WASTE SEPARATION, COLLECTION AND CONTAMINATION

CT-061 to CT-070 | 10 questions

8. QUALITY CHECKS, HANDOVER, WELLBEING AND LOWER-IMPACT OPERATIONS

CT-071 to CT-080 | 10 questions

Evidence precedence for operational answers

1
CURRENT AUTHORITATIVE RECORD

2
DIRECT OPERATIONAL OBSERVATION

3
PROPOSED TELEMETRY

4
DERIVED ANALYSIS

5
HISTORICAL SPATIAL REFERENCE

Conflict rule: do not average contradictory sources. Current responsible-owner records and competent direct observations within scope outrank weaker telemetry. Active safety/security restrictions use the more restrictive valid instruction until resolved. Permission-limited evidence remains permission-limited.

1. SHIFTS, ZONES, STANDARDS AND DAILY PRIORITIES

CT-001 What is my cleaning zone and task list for this shift, including anything added, removed or handed over?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor or contract manager for the shift rota and handover; building operations for live closures. Precedence: live restriction/closure notice, then current rota, then handover note. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: comparison.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: current rota plus latest handover/closure status.. Method: zone join and conflict check.. Uncertainty: flag stale or contradictory items.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer where individual workers are, score performance, reveal another person's assignment beyond need-to-know, grant access, or treat historical room labels as current. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Lets the team start in the right zones, avoid duplicated effort and immediately raise unclear or conflicting assignments.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-002 Which rooms must be ready first today, and what is the latest practical start time for each one?

Sensors and telemetry	Proposed: anonymous doorway counter or PIR/room-presence state at explicitly mapped room/zone boundaries; no camera, microphone, Wi-Fi/Bluetooth identifiers or person trajectories. Coverage: only commissioned rooms; sample 30-60 s and aggregate to 5-15 min room/zone counts or occupied/unoccupied/unknown state. Quality gate: commissioned asset/location mapping; validate detection/counting against periodic manual spot checks and document occlusion/blind spots; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for priorities; timetabling/event owner for booked use; local space owner for release. Precedence: explicit release/closure instruction over timetable, then verified telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + estimate + recommendation.. Interval: every 5-15 minutes during active room-turnaround or route decisions, and on release/booking change.. Minimum: current task list, next-use time and release status.. Method: deadline ordering with travel buffer.. Uncertainty: confidence band on vacancy.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not declare a room empty or safe from sensors alone, cancel bookings, infer who is present, or override local release rules and cleaning standards. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps sequence work so rooms are ready before use while reducing avoidable interruptions.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

1. SHIFTS, ZONES, STANDARDS AND DAILY PRIORITIES

CT-003 Are any of my normal areas closed, restricted or temporarily reassigned on this shift?

Sensors and telemetry	Proposed: anonymous doorway counter or PIR/room-presence state at explicitly mapped room/zone boundaries; no camera, microphone, Wi-Fi/Bluetooth identifiers or person trajectories. Coverage: only commissioned rooms; sample 30-60 s and aggregate to 5-15 min room/zone counts or occupied/unoccupied/unknown state. Quality gate: commissioned asset/location mapping; validate detection/counting against periodic manual spot checks and document occlusion/blind spots; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: security/building operations for restrictions, space owner for local closure, cleaning supervisor for reassignment. Precedence: current authorised closure/restriction notice over all sensed state. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: current restriction feed or supervisor instruction.. Method: role-based zone intersection.. Uncertainty: unknown where owner data is unavailable.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not expose sensitive security reasons or restricted-area details to unauthorised users, grant entry, or infer access from a door being open. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Prevents wasted travel and accidental entry into spaces that are currently unavailable or assigned elsewhere.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-004 What cleaning standard, frequency and task set apply to this room on this visit?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target ≤ 60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: cleaning contract/specification owner and supervisor for the current task standard; space owner may supply approved local constraints. Precedence: current contract/task specification and authorised local instruction. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: on demand and whenever the decisive evidence changes.. Minimum: verified room identity plus current standard/version.. Method: rule lookup by space type and service level.. Uncertainty: if room type or specification mapping is unresolved, safe failure is to ask the supervisor.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not invent a cleaning method, lower or raise required standards, select chemicals, or use the May 2021 plan to assert current room function. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Gives the operative the correct current task set without relying on memory or outdated room labels.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

1. SHIFTS, ZONES, STANDARDS AND DAILY PRIORITIES

CT-005 Which assigned tasks are now at real risk of missing their required completion time?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning work-order/shift system and supervisor. Precedence: supervisor-approved task status and explicit deferral over inferred progress. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + estimate + comparison + recommendation.. Interval: on demand and whenever the decisive evidence changes.. Minimum: task list, due times and explicit completion/defer states.. Method: deadline-risk rules using remaining standard task duration and travel allowance. Uncertainty is shown; safe failure avoids attributing cause. Decision interval: on demand and whenever the decisive evidence changes. Do not convert missing coverage into a zero/false value or hide conflicts to produce a cleaner ranking.. Uncertainty: show stale, missing, unmapped and conflicting evidence separately.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer individual productivity, blame a worker, auto-reassign work, or promise completion where access, release or resources remain uncertain. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Supports early escalation, realistic reprioritisation and a cleaner handover before deadlines are missed.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-006 Has an event, exam or other high-use period changed where we should add cleaning effort today?

Sensors and telemetry	Proposed: anonymous doorway/room occupancy aggregates plus fill-level sensors only on mapped bins/dispensers. Occupancy is 5-15 min aggregate; level sensors read every 5-15 min or on threshold change; no person history or content sensing. Quality gate: commissioned asset/location mapping; validate occupancy counts; calibrate each level sensor at known empty/full references and check post-refill drift; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: event/timetable system and cleaning supervisor for operational priorities; building operations for special instructions. Precedence: explicit event plan and supervisor direction over demand estimates. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: estimate + comparison + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: confirmed event or sustained aggregate demand signal from at least two intervals.. Method: threshold/baseline comparison.. Uncertainty: label sensor coverage gaps.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer event attendance by named people, identify groups, change staffing automatically, or replace the supervisor's decision on service levels. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Lets the team adapt routes and consumables planning when actual building demand differs from an ordinary day.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

1. SHIFTS, ZONES, STANDARDS AND DAILY PRIORITIES

CT-007 Which jobs were left unfinished or deliberately deferred by the previous shift and still need action?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: prior-shift handover and cleaning task system, with supervisor as current decision owner. Precedence: explicit handover/defer record over inferred status. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: unresolved prior task with a current status.. Method: deduplicate by room/task/time and check current accessibility.. Uncertainty: contradictory completion and defer records remain unresolved.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not treat absence of a completion scan as proof work was missed, expose individual performance histories, or silently close old tasks. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Makes unfinished work visible so it can be planned rather than discovered late in the shift.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-008 Where can I begin work now without interrupting teaching, meetings or another authorised activity?

Sensors and telemetry	Proposed: anonymous doorway counter or PIR/room-presence state at explicitly mapped room/zone boundaries; no camera, microphone, Wi-Fi/Bluetooth identifiers or person trajectories. Coverage: only commissioned rooms; sample 30-60 s and aggregate to 5-15 min room/zone counts or occupied/unoccupied/unknown state. Quality gate: commissioned asset/location mapping; validate detection/counting against periodic manual spot checks and document occlusion/blind spots; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: timetable/event owner and local space owner for formal use/release; cleaning supervisor for route choice. Precedence: explicit 'do not enter'/release instruction over timetable, then telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + recommendation.. Interval: every 5-15 minutes during active room-turnaround or route decisions, and on release/booking change. Safe failure: return an explicit unresolved state and the current responsible owner rather than guessing.. Minimum: current task plus confirmed access/release rule.. Method: filter out active/restricted spaces and rank by route distance.. Uncertainty: unknown release means exclude where confirmation is required.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not monitor speech, infer activity type from people, enter because a room seems quiet, or override booked-use and local access rules. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Reduces interruptions and helps use short cleaning windows productively.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

1. SHIFTS, ZONES, STANDARDS AND DAILY PRIORITIES

CT-009 Do entrances or circulation areas need an extra wet-weather check before floor conditions worsen?

Sensors and telemetry	Proposed: fixed conductivity/wetness strips only at commissioned entrance thresholds, mat wells or known plumbing risk points; optional authorised rain observation. These point sensors do not cover an entire floor. Wetness is event-driven or sampled ≤ 60 s; rain 5-10 min; use only the latest 10-30 min. Quality gate: commissioned asset/location mapping; wet/dry functional test; rain-gauge calibration where locally owned; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: building operations for weather-related instructions and cleaning supervisor for response; local weather feed is contextual. Precedence: observed hazard/report and authorised instruction over telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + comparison + recommendation.. Interval: on demand and whenever the decisive evidence changes.. Minimum: rainfall or wetness indication plus affected entrance mapping.. Method: simple rule/threshold.. Uncertainty: sensor false positives and unmapped entrances.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not certify a floor as safe, decide legal controls, deploy warning signs automatically, or replace local slip-prevention procedures. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps the team inspect and service the most affected entrances before water and dirt spread further.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-010 If we cannot finish everything, what exactly must be handed over to the next shift?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor and work-order/handover system. Precedence: confirmed task/fault status and current restrictions over inferred demand. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + calculation + recommendation.. Interval: on demand and whenever the decisive evidence changes.. Minimum: each item has location, status and last update.. Method: exception summary grouped by urgency/zone.. Uncertainty: distinguish unknown from not done.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not assign blame, infer why work was incomplete, conceal unresolved items, auto-authorise overtime or transfer restricted details to unauthorised staff. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Produces a concise, actionable handover so the next team knows what remains and what needs escalation.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

2. OCCUPANCY-DRIVEN DEMAND AND ROOM-RELEASE INFORMATION

CT-011 Which teaching or meeting rooms are formally released and ready for us to enter for cleaning now?

Sensors and telemetry	Proposed: anonymous doorway counter or PIR/room-presence state at explicitly mapped room/zone boundaries; no camera, microphone, Wi-Fi/Bluetooth identifiers or person trajectories. Coverage: only commissioned rooms; sample 30-60 s and aggregate to 5-15 min room/zone counts or occupied/unoccupied/unknown state. Quality gate: commissioned asset/location mapping; validate detection/counting against periodic manual spot checks and document occlusion/blind spots; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: timetabling/room-booking service and local space owner for release; cleaning supervisor for operational use. Precedence: explicit release or continued-use notice over timetable, then telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation.. Interval: every 5-15 minutes during active room-turnaround or route decisions, and on release/booking change.. Minimum: formal release state plus task need.. Method: rule-based eligibility.. Uncertainty: occupancy disagreement remains 'check/confirm'.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not identify occupants, infer that a booked group has left, declare a room empty from one sensor, or override release rules. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Lets cleaners act quickly when rooms genuinely become available without interrupting users.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-012 Which assigned rooms are likely to become cleanable in the next 30 to 60 minutes so I can plan the route?

Sensors and telemetry	Proposed: anonymous doorway counter or PIR/room-presence state at explicitly mapped room/zone boundaries; no camera, microphone, Wi-Fi/Bluetooth identifiers or person trajectories. Coverage: only commissioned rooms; sample 30-60 s and aggregate to 5-15 min room/zone counts or occupied/unoccupied/unknown state. Quality gate: commissioned asset/location mapping; validate detection/counting against periodic manual spot checks and document occlusion/blind spots; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: timetable/event owner for planned end times and local space owner for extensions; cleaning supervisor for route planning. Precedence: live extension/release notice over scheduled end, then telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: estimate + forecast + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: current booking plus reliable end time; telemetry only adjusts confidence.. Method: deterministic schedule with bounded delay estimate. Uncertainty shown as window; safe failure uses scheduled release only. Decision interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes. Do not convert missing coverage into a zero/false value or hide conflicts to produce a cleaner ranking.. Uncertainty: show stale, missing, unmapped and conflicting evidence separately.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not predict when a named person will leave, treat presence as consent to monitor, or promise room access at a forecast time. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps build an efficient route around likely cleaning windows instead of waiting at closed doors.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

2. OCCUPANCY-DRIVEN DEMAND AND ROOM-RELEASE INFORMATION

CT-013 Which common or study areas deserve an extra physical service check because recent use is above their normal pattern?

Sensors and telemetry	Proposed: anonymous doorway/room occupancy aggregates plus fill-level sensors only on mapped bins/dispensers. Occupancy is 5-15 min aggregate; level sensors read every 5-15 min or on threshold change; no person history or content sensing. Quality gate: commissioned asset/location mapping; validate occupancy counts; calibrate each level sensor at known empty/full references and check post-refill drift; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for service response; facilities/building operations for telemetry. Precedence: observed/report-based need and supervisor instruction over automated demand score. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: comparison + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: at least two healthy signals or one sustained signal over two intervals.. Method: baseline percentile/threshold.. Uncertainty: coverage gaps shown.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not identify users, infer demographic or behavioural traits, label people as causing mess, or change required cleaning standards automatically. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps send attention to areas where bins, touchpoints or general tidiness are likely to need it sooner.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-014 Has an event finished early enough to create a useful cleaning window before the next booking?

Sensors and telemetry	Proposed: anonymous doorway counter or PIR/room-presence state at explicitly mapped room/zone boundaries; no camera, microphone, Wi-Fi/Bluetooth identifiers or person trajectories. Coverage: only commissioned rooms; sample 30-60 s and aggregate to 5-15 min room/zone counts or occupied/unoccupied/unknown state. Quality gate: commissioned asset/location mapping; validate detection/counting against periodic manual spot checks and document occlusion/blind spots; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: event organiser/room owner for release and timetable system for schedule; cleaning supervisor for action. Precedence: explicit release over all inferred signals. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + calculation.. Interval: every 5-15 minutes during active room-turnaround or route decisions, and on release/booking change.. Minimum: booking still active plus sustained likely vacancy, then require release where policy says so.. Method: event-status exception flag. Uncertainty remains explicit; safe failure is 'possible window - confirm'. Decision interval: every 5-15 minutes during active room-turnaround or route decisions, and on release/booking change. Do not convert missing coverage into a zero/false value or hide conflicts to produce a cleaner ranking.. Uncertainty: show stale, missing, unmapped and conflicting evidence separately.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not contact or identify attendees, infer meeting content, cancel a booking, or treat an apparently empty room as released. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Creates useful extra work windows while respecting formal room control.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

2. OCCUPANCY-DRIVEN DEMAND AND ROOM-RELEASE INFORMATION

CT-015 The timetable says this room is free but the sensors suggest activity - what should I verify before entering?

Sensors and telemetry	Proposed: anonymous doorway counter or PIR/room-presence state at explicitly mapped room/zone boundaries; no camera, microphone, Wi-Fi/Bluetooth identifiers or person trajectories. Coverage: only commissioned rooms; sample 30-60 s and aggregate to 5-15 min room/zone counts or occupied/unoccupied/unknown state. Quality gate: commissioned asset/location mapping; validate detection/counting against periodic manual spot checks and document occlusion/blind spots; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: timetable system for planned state, local space owner/security for actual permitted use, cleaning supervisor for route. Precedence: current authorised local use/restriction over schedule. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + comparison.. Interval: every 5-15 minutes during active room-turnaround or route decisions, and on release/booking change.. Minimum: current free schedule plus two consistent presence intervals or a local report.. Method: conflict flag.. Uncertainty: possible sensor fault.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer who is inside or what they are doing, challenge occupants automatically, or treat occupancy telemetry as an access-control decision. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Prevents avoidable interruptions and highlights rooms whose actual use differs from planned use.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-016 If urgent work appears elsewhere, which assigned areas have genuine scheduling slack and could be serviced later?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for reprioritisation and current service specification for required frequencies. Precedence: mandatory scheduled tasks and explicit priorities over telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + estimate + comparison + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: task is still within allowed service window plus sustained low demand.. Method: bounded prioritisation, never below required frequency.. Uncertainty: unknown demand does not qualify.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not lower required service standards, skip a statutory/contractual task, infer cleanliness from low occupancy, or auto-defer work. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Gives the supervisor evidence to protect urgent work while keeping lower-demand tasks within their allowed windows.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

2. OCCUPANCY-DRIVEN DEMAND AND ROOM-RELEASE INFORMATION

CT-017 Where has recent room use increased enough to justify an extra cleaning or replenishment check?

Sensors and telemetry	Proposed: anonymous doorway/room occupancy aggregates plus fill-level sensors only on mapped bins/dispensers. Occupancy is 5-15 min aggregate; level sensors read every 5-15 min or on threshold change; no person history or content sensing. Quality gate: commissioned asset/location mapping; validate occupancy counts; calibrate each level sensor at known empty/full references and check post-refill drift; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for extra-pass decision; telemetry owner for sensor health; helpdesk for service calls. Precedence: direct hazard/service reports and supervisor judgement over demand model. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: estimate + comparison + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: sustained demand above agreed baseline or corroborating signals.. Method: percentile exceedance with recent-clean exclusion.. Uncertainty: seasonal/event effects.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not equate footfall with dirt, infer user behaviour or group characteristics, or automatically create mandatory work or staffing changes. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps target discretionary service passes where they are most likely to add value.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-018 Which rooms have an uncertain release status, and who or what is the right current source to confirm each one?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target ≤ 60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: room owner/event organiser for release, security/building operations for restrictions, cleaning supervisor for workflow. Precedence: authorised owner response over timetable or telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: every 5-15 minutes during active room-turnaround or route decisions, and on release/booking change.. Minimum: task exists but release cannot be established.. Method: conflict classification and role lookup. Uncertainty is the result; safe failure directs to authorised contact rather than guessing. Decision interval: every 5-15 minutes during active room-turnaround or route decisions, and on release/booking change. Do not convert missing coverage into a zero/false value or hide conflicts to produce a cleaner ranking.. Uncertainty: show stale, missing, unmapped and conflicting evidence separately.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not disclose private contact details unnecessarily, infer consent from open doors, or grant access. Only role-appropriate contact channels should be shown. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Lets the team resolve uncertainty quickly instead of repeatedly checking a room or taking an unsafe assumption.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

2. OCCUPANCY-DRIVEN DEMAND AND ROOM-RELEASE INFORMATION

CT-019 When is the least disruptive window for cleaning a busy corridor, staircase or lift lobby?

Sensors and telemetry	Proposed: anonymous doorway counter or PIR/room-presence state at explicitly mapped room/zone boundaries; no camera, microphone, Wi-Fi/Bluetooth identifiers or person trajectories. Coverage: only commissioned rooms; sample 30-60 s and aggregate to 5-15 min room/zone counts or occupied/unoccupied/unknown state. Quality gate: commissioned asset/location mapping; validate detection/counting against periodic manual spot checks and document occlusion/blind spots; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for timing, building operations/security for restrictions, facilities team for sensor data. Precedence: emergency/access restrictions and scheduled high-priority movements over traffic estimates. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: estimate + comparison + recommendation.. Interval: every 5-15 minutes during active room-turnaround or route decisions, and on release/booking change.. Minimum: at least three healthy recent intervals plus no overriding restriction.. Method: rolling minimum forecast from recent pattern.. Uncertainty: event surges may break pattern.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not track individuals, block egress, authorise barriers, or certify that a route will remain clear. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps time floor work and trolley movement to reduce obstruction and user disruption.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-020 After a large event or teaching release, which public areas should we physically check first?

Sensors and telemetry	Proposed: anonymous doorway/room occupancy aggregates plus fill-level sensors only on mapped bins/dispensers. Occupancy is 5-15 min aggregate; level sensors read every 5-15 min or on threshold change; no person history or content sensing. Quality gate: commissioned asset/location mapping; validate occupancy counts; calibrate each level sensor at known empty/full references and check post-refill drift; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: event plan and cleaning supervisor for response; facilities telemetry owner for sensed demand. Precedence: reported spill/hazard and event-specific cleaning plan over aggregate score. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + comparison + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: event footprint plus at least one reliable demand signal per area.. Method: weighted demand score with explicit factors.. Uncertainty: partial coverage lowers confidence.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer attendee behaviour, identify groups, inspect event content or replace the approved post-event cleaning plan. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Focuses the first inspection on areas most likely to need bins, washrooms or general tidiness attended quickly.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

3. WASHROOMS, CONSUMABLES AND AMENITIES

CT-021 Which washroom should I service next, considering required frequency, time since the last confirmed service and current demand evidence?

Sensors and telemetry	Proposed: anonymous doorway/room occupancy aggregates plus fill-level sensors only on mapped bins/dispensers. Occupancy is 5-15 min aggregate; level sensors read every 5-15 min or on threshold change; no person history or content sensing. Quality gate: commissioned asset/location mapping; validate occupancy counts; calibrate each level sensor at known empty/full references and check post-refill drift; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning task system and supervisor for service status; facilities for telemetry. Precedence: direct user/fault report and required cleaning frequency over demand estimate. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + calculation + comparison + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: current due window plus sustained demand or consumable/bin signal.. Method: rule-based score.. Uncertainty: doorway count cannot prove condition.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer who used a washroom, protected traits, personal routines or hygiene behaviour, and must not reduce required service frequency. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps choose the next washroom check when several are due around the same time.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-022 Which soap, paper or other approved dispensers are likely to be low or empty before I start the route?

Sensors and telemetry	Proposed: load-cell, time-of-flight or optical-distance level sensors on specific mapped dispensers/bins; no camera-based estimation. Read every 5-15 min or threshold event, with latest level and short rate-of-change only. Quality gate: commissioned asset/location mapping; empty/full reference calibration for each asset, refill/empty reset checks and drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning/facilities inventory system for dispenser mapping and supervisor for refill task; sensor platform for telemetry. Precedence: physical check and explicit refill record over sensor estimate. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + estimate + forecast.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: healthy recent reading below configured level or confirmed manual report.. Method: threshold with hysteresis.. Uncertainty: stale/implausible readings become 'check manually'.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer individual usage, expose washroom-use patterns beyond operational aggregation, or order supplies automatically without authorised workflow. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Lets the operative load the right consumables and avoid repeated store trips.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

3. WASHROOMS, CONSUMABLES AND AMENITIES

CT-023 Are any washrooms, cubicles or amenity points officially out of service before I travel there?

Sensors and telemetry	Proposed: door contact/controller health or access-denied/accepted event only for the mapped door and only at role-permitted detail; never reconstruct a person's route. Events are immediate, with current controller heartbeat. Door state can show open/closed or a recorded controller fault, not permission or cause. Quality gate: commissioned asset/location mapping; door-contact/controller functional verification; credential entitlement remains outside telemetry; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: facilities/building operations for closure, cleaning supervisor for route, maintenance/helpdesk for faults. Precedence: current closure instruction and onsite signage over telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: active authorised closure or fault affecting service.. Method: status join by asset/location.. Uncertainty: unknown fault impact remains 'confirm'.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not reopen a facility, override signage, infer privacy-sensitive occupancy, or disclose maintenance details beyond role need. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Avoids wasted visits and helps route cleaning around facilities that cannot currently be serviced normally.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-024 Which shared amenity points, such as kitchenettes or drinking-water areas, need a service check today?

Sensors and telemetry	Proposed: anonymous doorway/room occupancy aggregates plus fill-level sensors only on mapped bins/dispensers. Occupancy is 5-15 min aggregate; level sensors read every 5-15 min or on threshold change; no person history or content sensing. Quality gate: commissioned asset/location mapping; validate occupancy counts; calibrate each level sensor at known empty/full references and check post-refill drift; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for service tasks, facilities for asset status, water/technical competent owner for water-quality or hygiene controls. Precedence: formal service standard and fault/closure notice over usage estimate. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + comparison + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: mapped asset and current service requirement.. Method: due-task plus use/fault flag.. Uncertainty: sensor absence means no demand inference.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not certify water quality, diagnose plumbing, change hygiene controls, or infer personal consumption patterns. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps include busy or fault-affected amenity points in the day's route without missing required scheduled care.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

3. WASHROOMS, CONSUMABLES AND AMENITIES

CT-025 Which washrooms have a confirmed service this shift, and which still need a visit?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning task system and supervisor. Precedence: verified completion/defer record over telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: current shift task roster and explicit states.. Method: count by location/status, excluding closed facilities.. Uncertainty: missing scan is 'unknown' unless policy defines otherwise.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer individual productivity, treat sensor silence as completion, or expose named performance statistics. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Gives the team a reliable route-completion picture and makes omissions visible before handover.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-026 Which washrooms regularly need more consumables so we can plan stock for the coming week?

Sensors and telemetry	Proposed: load-cell, time-of-flight or optical-distance level sensors on specific mapped dispensers/bins; no camera-based estimation. Read every 5-15 min or threshold event, with latest level and short rate-of-change only. Quality gate: commissioned asset/location mapping; empty/full reference calibration for each asset, refill/empty reset checks and drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: inventory/cleaning supervisor for refill records, facilities for telemetry, procurement/stores for stock availability. Precedence: counted stock movement and confirmed refills over sensor-estimated consumption. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + estimate + forecast + comparison.. Interval: daily or weekly planning assessment, while current urgent faults remain event-driven.. Minimum: at least two comparable weeks with sufficient refill coverage.. Method: median/quantile demand by washroom and day type.. Uncertainty: event/term changes and missing refills.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer user identities or protected traits from washroom demand, set stock policy automatically, or conceal data-quality gaps. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Supports better trolley and store stocking so high-demand locations are less likely to run out.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

3. WASHROOMS, CONSUMABLES AND AMENITIES

CT-027 Where do dispenser or bin sensor readings disagree with what staff actually found on the last physical check?

Sensors and telemetry	Proposed: load-cell, time-of-flight or optical-distance level sensors on specific mapped dispensers/bins; no camera-based estimation. Read every 5-15 min or threshold event, with latest level and short rate-of-change only. Quality gate: commissioned asset/location mapping; empty/full reference calibration for each asset, refill/empty reset checks and drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: sensor platform owner for readings and cleaning supervisor for manual records. Precedence: current physical check for operational action, while telemetry remains diagnostic. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + comparison + diagnosis.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: paired sensor and manual observation within an agreed time window.. Method: tolerance comparison; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: intervening use/refill may explain mismatch.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not accuse staff of incorrect reporting, auto-correct records without traceability, or use discrepancies for performance scoring. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps identify unreliable sensors so staff do not waste trips based on bad low-stock alerts.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-028 Is there a current leak or standing-water alert in a washroom that needs escalation before routine cleaning?

Sensors and telemetry	Proposed: fixed leak rope/point conductivity sensor near mapped plumbing/plant risk points and, where exposed, BMS water/leak alarm points. Coverage: limited to instrumented points/zones; alarm event-driven with 1-5 min heartbeat. Quality gate: commissioned asset/location mapping; wet/dry functional test plus BMS point/alarm-limit validation by the owning service; clocks aligned to building time (target ≤ 30 s); uptime $\geq 95\%$ for an urgent alarm window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: building operations/maintenance for leak alarms and isolation decisions; cleaning supervisor for cleaning response. Precedence: emergency/facilities instruction and physical observation over telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: healthy leak/wetness alarm or confirmed report.. Method: alarm validation against flow/status where available.. Uncertainty: false alarms possible.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not direct plumbing isolation, certify electrical or slip safety, identify the leak source beyond evidence, or send staff into a hazardous area contrary to procedures. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Enables rapid escalation and helps cleaners avoid treating a continuing technical leak as an ordinary mopping task.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

3. WASHROOMS, CONSUMABLES AND AMENITIES

CT-029 Do accessible or assisted-use facilities have any current service issue I need to know about before the round?

Sensors and telemetry	Proposed: anonymous doorway/room occupancy aggregates plus fill-level sensors only on mapped bins/dispensers. Occupancy is 5-15 min aggregate; level sensors read every 5-15 min or on threshold change; no person history or content sensing. Quality gate: commissioned asset/location mapping; validate occupancy counts; calibrate each level sensor at known empty/full references and check post-refill drift; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: facilities/accessibility service owner for facility status, cleaning supervisor for service, maintenance/helpdesk for faults. Precedence: current authorised closure/fault instruction over telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: on demand and whenever the decisive evidence changes.. Minimum: mapped facility plus current service/fault state.. Method: simple exception list.. Uncertainty: missing telemetry does not imply normal operation.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer who uses the facility, disclose protected traits, or claim compliance/accessibility certification from cleaning data. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps ensure these facilities receive timely service and that known faults are not mistaken for cleaning issues.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-030 How much approved washroom stock should I take on this route to reduce avoidable return trips?

Sensors and telemetry	Proposed: anonymous doorway/room occupancy aggregates plus fill-level sensors only on mapped bins/dispensers. Occupancy is 5-15 min aggregate; level sensors read every 5-15 min or on threshold change; no person history or content sensing. Quality gate: commissioned asset/location mapping; validate occupancy counts; calibrate each level sensor at known empty/full references and check post-refill drift; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: stores/inventory owner for stock counts, cleaning supervisor for route plan, procurement for replenishment status. Precedence: verified physical stock and current route/task list over forecast. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + estimate + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: number/type of dispensers, current stock and recent refill history.. Method: upper-quantile demand plus small buffer within local policy.. Uncertainty: show range.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not authorise substitution of products, change stock controls, infer individual usage, or promise stock availability beyond the recorded store state. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Reduces return trips while avoiding excessive trolley loading and unnecessary stock movement.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

4. SPILLS, DAMAGE, FAULTS AND URGENT ESCALATION

CT-031 A spill has just been reported - where is it, when was it reported, and who currently owns the response?

Sensors and telemetry	Proposed: the time-stamped incident/service-desk report and acknowledgement workflow; arbitrary spills, glass, sharps or contamination should not be 'detected' by speculative computer vision. A fixed wetness/alarm sensor may add evidence only at its commissioned point. Event-driven reporting; no continuous person sensing. Quality gate: commissioned asset/location mapping; workflow event integrity; any fixed alarm requires its modality-specific functional test; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: helpdesk/building operations for incident record, cleaning supervisor for response, security/competent team for hazardous cases. Precedence: direct authorised incident report and onsite assessment over sensor alert. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation.. Interval: immediately on the incident, alarm or authoritative status change.. Minimum: a valid location plus report/healthy alarm.. Method: map to nearest verified service zone and classify only by reported type.. Uncertainty: ambiguous location prompts confirmation.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not identify the person who caused/reported the spill unless strictly needed, classify an unknown substance, certify the area safe, or prescribe a chemical response. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Gets the team to the right place quickly while preserving the correct escalation path for uncertain hazards.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-032 Has this reported spill already been attended, or is it still an open action for our team?

Sensors and telemetry	Proposed: the time-stamped incident/service-desk report and acknowledgement workflow; arbitrary spills, glass, sharps or contamination should not be 'detected' by speculative computer vision. A fixed wetness/alarm sensor may add evidence only at its commissioned point. Event-driven reporting; no continuous person sensing. Quality gate: commissioned asset/location mapping; workflow event integrity; any fixed alarm requires its modality-specific functional test; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor/helpdesk for incident state and building operations for any safety restriction. Precedence: explicit closure/completion and physical confirmation over telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison.. Interval: immediately on the incident, alarm or authoritative status change.. Minimum: incident ID/location plus latest authorised status.. Method: status join with optional sensor corroboration.. Uncertainty: unresolved contradiction remains open.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not close a safety incident automatically, certify dryness or slip safety, or infer completion from a falling moisture reading alone. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Prevents duplicate trips while ensuring potentially unresolved spills are not silently dismissed.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

4. SPILLS, DAMAGE, FAULTS AND URGENT ESCALATION

CT-033 Has broken glass, a sharp object or another physical hazard been reported in my area?

Sensors and telemetry	Proposed: the time-stamped incident/service-desk report and acknowledgement workflow; arbitrary spills, glass, sharps or contamination should not be 'detected' by speculative computer vision. A fixed wetness/alarm sensor may add evidence only at its commissioned point. Event-driven reporting; no continuous person sensing. Quality gate: commissioned asset/location mapping; workflow event integrity; any fixed alarm requires its modality-specific functional test; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: security/building operations and cleaning supervisor for incident handling; competent safety owner where required. Precedence: direct report/onsite assessment over any indirect sensor context. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation.. Interval: immediately on the incident, alarm or authoritative status change.. Minimum: authorised report with location and current state.. Method: filter by assigned zone and open status.. Uncertainty: unknown details remain unknown.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not tell staff to handle an unknown sharp object, certify an area clear, identify who caused it, or replace approved PPE/handling procedures. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Warns the team before entry so they can follow the correct safe-work and escalation process.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-034 This fixture or surface is damaged - is it within cleaning responsibility, or should I raise it to facilities?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target ≤ 60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: facilities/helpdesk for asset fault ownership, cleaning supervisor for cleaning task boundary. Precedence: current asset classification and competent technical instruction over sensor inference. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison + diagnosis + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: verified location/asset and observable issue description.. Method: bounded rules for 'cleaning task', 'maintenance task' or 'unknown - escalate'. Uncertainty is explicit; safe failure sends unclear cases to facilities. Decision interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable. Do not convert missing coverage into a zero/false value or hide conflicts to produce a cleaner ranking; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: show stale, missing, unmapped and conflicting evidence separately.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not diagnose technical failure, authorise maintenance work, assign blame, or instruct cleaning staff to alter damaged building systems. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Stops cleaners spending time on faults outside their role and gets technical problems to the correct owner faster.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

4. SPILLS, DAMAGE, FAULTS AND URGENT ESCALATION

CT-035 Is there a leak, flooding or plant alarm that changes whether normal cleaning can continue in this area?

Sensors and telemetry	Proposed: fixed leak rope/point conductivity sensor near mapped plumbing/plant risk points and, where exposed, BMS water/leak alarm points. Coverage: limited to instrumented points/zones; alarm event-driven with 1-5 min heartbeat. Quality gate: commissioned asset/location mapping; wet/dry functional test plus BMS point/alarm-limit validation by the owning service; clocks aligned to building time (target <=30 s); uptime >=95% for an urgent alarm window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: building operations/maintenance for leak response and safe access; cleaning supervisor for task changes. Precedence: emergency/technical instruction and physical assessment over sensor data. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/<=5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation + diagnosis + recommendation.. Interval: immediately on the incident, alarm or authoritative status change.. Minimum: active validated alarm or confirmed report.. Method: map affected zone and suppress normal cleaning recommendation until owner status known; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: treat uncertain active alarms cautiously.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not tell staff to enter floodwater, isolate utilities, diagnose source, certify area safety or resume work without authorised clearance. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps prevent an ordinary cleaning response from exposing staff to a continuing technical incident.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-036 A bin, sanitary unit or dispenser is damaged - is there already a fault logged, and what information should I add if not?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target <=60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: facilities/helpdesk for asset faults, cleaning supervisor for temporary service response, specialist supplier where contracted. Precedence: confirmed fault assessment over sensor anomaly. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/<=5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison + diagnosis.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: asset ID/location and confirmed abnormal condition or persistent sensor anomaly.. Method: asset-state rules; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: ambiguous anomaly prompts physical check.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not attribute damage to a person, dismantle equipment, authorise technical work or claim the unit is safe to use. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Allows the cleaner to separate a service task from a maintenance fault and avoid repeated ineffective cleaning visits.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

4. SPILLS, DAMAGE, FAULTS AND URGENT ESCALATION

CT-037 Which urgent cleaning-related incidents are still open and waiting for acknowledgement or action?

Sensors and telemetry	Proposed: the time-stamped incident/service-desk report and acknowledgement workflow; arbitrary spills, glass, sharps or contamination should not be 'detected' by speculative computer vision. A fixed wetness/alarm sensor may add evidence only at its commissioned point. Event-driven reporting; no continuous person sensing. Quality gate: commissioned asset/location mapping; workflow event integrity; any fixed alarm requires its modality-specific functional test; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: helpdesk/building operations for incident lifecycle and cleaning supervisor for cleaning action. Precedence: explicit acknowledgement/closure record over inferred activity. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison + recommendation.. Interval: immediately on the incident, alarm or authoritative status change.. Minimum: current open incident with cleaning relevance and urgency classification from owner.. Method: filter/sort by owner-defined urgency and age.. Uncertainty: missing acknowledgement remains open.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer individual responsiveness, score staff, change incident urgency or close incidents automatically. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Ensures high-priority incidents remain visible until someone authorised has taken ownership.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-038 There is an odour complaint - what verified information do we have before I investigate or escalate it?

Sensors and telemetry	Proposed: mapped temperature/humidity and, only where legitimately installed, broad IAQ/VOC trend sensors; no audio. Use 1-5 min samples aggregated to 15-min zone trends around the report. Generic IAQ cannot identify an odour or substance. Quality gate: commissioned asset/location mapping; calibration appropriate to each environmental sensor and explicit cross-sensitivity/drift checks; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: helpdesk for complaint, building operations for ventilation/technical context, cleaning supervisor for first response. Precedence: direct onsite assessment and competent technical/safety advice over VOC or environmental telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + comparison + diagnosis.. Interval: immediately on the incident, alarm or authoritative status change.. Minimum: location/time and current area status.. Method: correlate environmental anomalies within a narrow window without causal claims; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: many possible sources.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not identify a chemical from generic VOC data, infer personal hygiene or behaviour, monitor conversations, or prescribe masking/chemical treatment. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Gives staff useful context while avoiding false certainty about a smell that may need facilities or safety escalation.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

4. SPILLS, DAMAGE, FAULTS AND URGENT ESCALATION

CT-039 Do entrance floors show enough wet-weather evidence to justify an urgent physical check?

Sensors and telemetry	Proposed: fixed conductivity/wetness strips only at commissioned entrance thresholds, mat wells or known plumbing risk points; optional authorised rain observation. These point sensors do not cover an entire floor. Wetness is event-driven or sampled ≤ 60 s; rain 5-10 min; use only the latest 10-30 min. Quality gate: commissioned asset/location mapping; wet/dry functional test; rain-gauge calibration where locally owned; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for response, building operations for weather/hazard instructions. Precedence: physical observation/report and authorised control measures over sensor score. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + comparison + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: rainfall plus entrance moisture/wetness or confirmed report.. Method: simple combined threshold.. Uncertainty: local tracking varies.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not certify slip safety, block routes, change emergency egress arrangements or automatically deploy controls. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Prompts timely entrance checks that can contain tracked-in water and dirt before it spreads.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-040 If I find an unknown or suspected hazardous substance, what is the current escalation route and what facts should I pass on?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target ≤ 60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: current safety/COSHH management, building operations/security and cleaning supervisor. Precedence: current emergency/COSHH procedure and competent professional instruction over any digital suggestion. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: immediately on the incident, alarm or authoritative status change.. Minimum: verified building/zone and current procedure version.. Method: role-based document/contact lookup only.. Uncertainty: if procedure/version cannot be verified, safe failure is to stop guidance and direct to supervisor/emergency route.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not identify the substance, prescribe containment or chemical response, advise handling, certify PPE, or replace COSHH and emergency procedures. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Lets staff move quickly to the correct competent owner without turning the assistant into a hazardous-substance decision maker.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

5. KEYS, ACCESS, LONE WORKING AND SECURITY BOUNDARIES

CT-041 Which doors and service zones is my current role authorised to access for this shift?

Sensors and telemetry	Proposed: door contact/controller health or access-denied/accepted event only for the mapped door and only at role-permitted detail; never reconstruct a person's route. Events are immediate, with current controller heartbeat. Door state can show open/closed or a recorded controller fault, not permission or cause. Quality gate: commissioned asset/location mapping; door-contact/controller functional verification; credential entitlement remains outside telemetry; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: security/access-control administrator for entitlements, cleaning supervisor for work zones. Precedence: current access-control policy and explicit restriction over rota or historical mapping. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: authenticated role/credential context plus current entitlement record.. Method: allow-list intersection only.. Uncertainty: missing/expired entitlement becomes 'not confirmed'.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not grant, extend or bypass access, reveal sensitive zone details to unauthorised users, or infer permission from previous badge use or an open door. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Clarifies the authorised work boundary before the route begins and reduces avoidable access problems.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-042 For this task, do I need a key, access card, escort, permit or local contact before I travel there?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target ≤ 60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: security/building operations for access mode, permit owner for permits, cleaning supervisor for task. Precedence: current security/permit rule and local restriction over old room data. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: verified destination and current task.. Method: rule lookup by role, zone and time.. Uncertainty: unknown requirement prompts contact with security/supervisor.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not issue keys/cards/permits, book an escort without authorised workflow, disclose secure procedures broadly, or advise bypassing controls. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Lets the cleaner prepare the right access arrangement before arriving at a controlled space.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

5. KEYS, ACCESS, LONE WORKING AND SECURITY BOUNDARIES

CT-043 A room is locked even though my task says it should be available - is there a current known reason?

Sensors and telemetry	Proposed: door contact/controller health or access-denied/accepted event only for the mapped door and only at role-permitted detail; never reconstruct a person's route. Events are immediate, with current controller heartbeat. Door state can show open/closed or a recorded controller fault, not permission or cause. Quality gate: commissioned asset/location mapping; door-contact/controller functional verification; credential entitlement remains outside telemetry; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: security/building operations for door/access status, room owner for local restriction, cleaning supervisor for task. Precedence: current restriction/security status over schedule. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation + comparison + diagnosis.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: verified room plus current restriction/fault record.. Method: status lookup without person attribution; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: no known reason is reported as unknown.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not reveal access-log identities, remotely unlock doors, suggest tailgating/bypass, or infer security intent from a lock state. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Saves time by distinguishing a known restriction or door fault from a simple route mismatch.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-044 What authorised route can I use to reach this work area without passing through restricted research or secure zones?

Sensors and telemetry	Proposed: current mapped route/door closure state plus, only as secondary context, anonymous corridor-flow counts aggregated by zone; no trajectories. Restriction events are live; flow is 5-min aggregate over the last 30-60 min. Quality gate: commissioned asset/location mapping; current route-edge/door mapping verification and manual validation of any aggregate flow counter; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: security/building operations for authorised routes and restrictions, cleaning supervisor for destination. Precedence: current access/closure map over the May 2021 plan. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + comparison + recommendation.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: verified origin/destination plus current allowed-zone graph.. Method: shortest path excluding restricted/closed edges.. Uncertainty: unverified connection is excluded.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not reveal sensitive secure-area topology beyond need-to-know, route through emergency-only doors, or treat a historical plan as current access authority. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Provides a practical route that respects access boundaries and reduces failed attempts at controlled doors.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

5. KEYS, ACCESS, LONE WORKING AND SECURITY BOUNDARIES

CT-045 Does this location or task have a current lone-working check-in or contact requirement for this shift?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target <=60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: employer safety/lone-working policy owner and cleaning supervisor; security may own check-in channel. Precedence: current risk assessment/lone-working procedure over system defaults. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/<=5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: verified task/location and current policy/risk record.. Method: rules lookup.. Uncertainty: if mapping or document validity is unclear, safe failure directs to supervisor and does not declare lone working acceptable.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not monitor a worker continuously, infer wellbeing from movement, certify a task safe for lone working, or replace risk assessment and supervisor judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Reminds staff of required check-in or accompaniment arrangements before starting a relevant task.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-046 Will any time-limited access, permit or escort arrangement expire before this planned route is finished?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target <=60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: security/access-control administrator for entitlement validity and cleaning supervisor for shift timing. Precedence: current entitlement record over rota assumptions. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/<=5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: authenticated entitlement plus planned end time.. Method: time-window intersection.. Uncertainty: access-system outage or pending change.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not extend credentials, promise future access, reveal security policy beyond role need, or use past entries to infer permission. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Lets the team resolve an expiring credential before it causes a failed visit late in the shift.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

5. KEYS, ACCESS, LONE WORKING AND SECURITY BOUNDARIES

CT-047 Which rooms on my list require an escort, responsible person or local contact before cleaning can begin?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target <=60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: security/space owner for local access conditions, cleaning supervisor for task. Precedence: current room restriction and permit rule over any routine schedule. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/<=5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: verified room and current access condition.. Method: role/rule lookup.. Uncertainty: missing condition data means 'confirm before entry'.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not identify or locate a particular person, infer that an escort is present, waive the requirement or grant access. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Prevents accidental entry into controlled rooms and makes required coordination visible in advance.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-048 If I cannot get into a room, who currently owns the decision to release it or arrange access?

Sensors and telemetry	Proposed: door contact/controller health or access-denied/accepted event only for the mapped door and only at role-permitted detail; never reconstruct a person's route. Events are immediate, with current controller heartbeat. Door state can show open/closed or a recorded controller fault, not permission or cause. Quality gate: commissioned asset/location mapping; door-contact/controller functional verification; credential entitlement remains outside telemetry; clock alignment target <=60 s; uptime >=80% of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: space-management directory for room owner role, security for access authority, cleaning supervisor for task. Precedence: current authoritative owner/role directory and security policy. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/<=5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison + diagnosis.. Interval: at shift/task start and whenever the decisive owner-controlled status changes. Safe failure: return an explicit unresolved state and the current responsible owner rather than guessing.. Minimum: verified room/zone and type of access problem.. Method: route to responsible role, not guessed individual; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: conflicting owners are shown and escalated to supervisor/security.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not grant access, disclose unnecessary personal details, identify previous occupants or suggest bypassing the lock/security process. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Gets access problems to the right decision owner without cleaners searching through informal contacts.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

5. KEYS, ACCESS, LONE WORKING AND SECURITY BOUNDARIES

CT-049 Is there a current security, emergency or fire-route restriction that changes my cleaning or waste route?

Sensors and telemetry	Proposed: current mapped route/door closure state plus, only as secondary context, anonymous corridor-flow counts aggregated by zone; no trajectories. Restriction events are live; flow is 5-min aggregate over the last 30-60 min. Quality gate: commissioned asset/location mapping; current route-edge/door mapping verification and manual validation of any aggregate flow counter; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: security/emergency management and building operations. Precedence: live authorised emergency/security instruction over all schedules, telemetry and historic plans. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: immediately on the incident, alarm or authoritative status change.. Minimum: active restriction relevant to route.. Method: remove affected zones/edges and show official instruction.. Uncertainty: if feed freshness cannot be verified, safe failure defers to alarms, signage and staff instructions.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not expose sensitive incident details, override emergency directions, route through blocked areas or control doors/alarms. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps staff alter routine work promptly while following the current security or emergency instruction.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-050 Where must keys, radios or other controlled service items be returned at the end of this shift?

Sensors and telemetry	Proposed: door contact/controller health or access-denied/accepted event only for the mapped door and only at role-permitted detail; never reconstruct a person's route. Events are immediate, with current controller heartbeat. Door state can show open/closed or a recorded controller fault, not permission or cause. Quality gate: commissioned asset/location mapping; door-contact/controller functional verification; credential entitlement remains outside telemetry; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: security/key-control or equipment custodian, with cleaning supervisor for shift handover. Precedence: current custody procedure and return assignment over habit or historical location. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: verified item and authorised user role.. Method: custody-rule lookup.. Uncertainty: unclear assignment prompts handover to supervisor/security.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not reveal key-store locations or codes to unauthorised users, transfer custody automatically, or mark an item returned without an explicit handover record. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Reduces lost keys/equipment and supports a clean, traceable end-of-shift handover.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

6. EQUIPMENT, CHEMICALS, STORAGE AND REPLENISHMENT

CT-051 Which cleaning machines and tools are currently available and recorded as serviceable for this route?

Sensors and telemetry	Proposed: the specific cleaning machine controller/BMS state, battery SoC/voltage/current, charger/dock state or hour counter where available; no staff location. Read 1-5 min or controller events and use only the current task/recent discharge window. Quality gate: commissioned asset/location mapping; controller point validation, SoC/energy functional checks and battery-age/drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning equipment custodian/supervisor for availability, maintenance provider for service status. Precedence: current inspection/out-of-service tag and supervisor instruction over telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison + recommendation.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: asset ID, current custody/availability and valid service-status record.. Method: filter by equipment type and status.. Uncertainty: missing inspection/health data becomes 'do not rely on digital status'.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not certify equipment safe, allocate it automatically, track staff through tagged equipment, or override pre-use checks and maintenance restrictions. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps staff choose an available machine without walking to equipment that is already in use or out of service.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-052 Where is the nearest authorised store that holds the approved supplies for my planned route?

Sensors and telemetry	Proposed: load-cell/smart-bin/smart-cabinet count for exact approved SKU at an authorised store; no staff tracking. Read 15-60 min and on issue/return where supported; aggregate by SKU/store. Quality gate: commissioned asset/location mapping; reference-weight/package-count calibration, cycle-count reconciliation and drift checks; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning/stores manager for store location and stock, security for access classification. Precedence: current authorised store register and access rules over historical plans. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + comparison + recommendation.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: verified need, current store record and permitted route.. Method: shortest authorised path.. Uncertainty: unverified store/route is excluded.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not disclose security-sensitive stores, unlock doors, reveal access codes or rely on May 2021 room use as current storage truth. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Cuts unnecessary trolley travel and directs staff only to stores they are allowed to use.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

6. EQUIPMENT, CHEMICALS, STORAGE AND REPLENISHMENT

CT-053 Does the floor-cleaning machine have enough verified battery charge for the planned job without a mid-task stop?

Sensors and telemetry	Proposed: the specific cleaning machine controller/BMS state, battery SoC/voltage/current, charger/dock state or hour counter where available; no staff location. Read 1-5 min or controller events and use only the current task/recent discharge window. Quality gate: commissioned asset/location mapping; controller point validation, SoC/energy functional checks and battery-age/drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: equipment custodian for readiness, maintenance provider for battery health, cleaning supervisor for route. Precedence: physical pre-use check and out-of-service status over telemetry estimate. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + estimate + forecast + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: recent state-of-charge plus reliable recent runtime data for that machine.. Method: conservative consumption estimate with reserve.. Uncertainty: report range.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not certify battery or equipment safety, override charging instructions, or promise completion of the route from a battery estimate. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps avoid a machine stopping halfway through a route and supports sensible equipment choice.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-054 Is any machine or powered tool due an inspection, service or safety check before I use it?

Sensors and telemetry	Proposed: the specific cleaning machine controller/BMS state, battery SoC/voltage/current, charger/dock state or hour counter where available; no staff location. Read 1-5 min or controller events and use only the current task/recent discharge window. Quality gate: commissioned asset/location mapping; controller point validation, SoC/energy functional checks and battery-age/drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: equipment custodian and competent maintenance provider. Precedence: current inspection/maintenance record and physical tag over sensor/runtime data. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + comparison.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: verified asset and current maintenance record.. Method: date/status rule.. Uncertainty: missing or conflicting record means status unverified.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not certify legal compliance or safety, extend an inspection date, clear equipment for use or replace pre-use checks. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Prevents staff choosing equipment whose required check is due, expired or unresolved.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

6. EQUIPMENT, CHEMICALS, STORAGE AND REPLENISHMENT

CT-055 Are the approved cleaning products I need actually in stock at the store I am allowed to use?

Sensors and telemetry	Proposed: load-cell/smart-bin/smart-cabinet count for exact approved SKU at an authorised store; no staff tracking. Read 15-60 min and on issue/return where supported; aggregate by SKU/store. Quality gate: commissioned asset/location mapping; reference-weight/package-count calibration, cycle-count reconciliation and drift checks; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: stores/inventory owner for stock, cleaning/COSHH owner for approved product list. Precedence: current approved-product register and physical stock check over sensor estimate. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: current task/product code and current inventory state.. Method: exact code match and quantity threshold.. Uncertainty: stale stock becomes 'check store'.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not recommend an unapproved substitute, authorise chemical use, infer container contents, or bypass COSHH/SDS requirements. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps staff confirm needed products are available before travelling to the task area.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-056 For this surface and task, which current approved product or method does the controlled cleaning specification require?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target ≤ 60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: current cleaning specification/COSHH owner and manufacturer/asset documentation where relevant. Precedence: approved task method, COSHH/SDS and competent supervisor instruction over all inferred material data. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: verified task and surface/material mapping.. Method: exact rules/document lookup only.. Uncertainty: unresolved surface/product mapping triggers supervisor confirmation.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not prescribe a chemical, invent dilution, recommend substitution, identify unknown residues or replace COSHH/SDS and training. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Makes the current approved instruction easy to find while keeping chemical decisions with the proper authority.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

6. EQUIPMENT, CHEMICALS, STORAGE AND REPLENISHMENT

CT-057 Are any chemical containers recorded as expired, damaged, unlabelled or otherwise not ready for issue?

Sensors and telemetry	Proposed: load-cell/smart-bin/smart-cabinet count for exact approved SKU at an authorised store; no staff tracking. Read 15-60 min and on issue/return where supported; aggregate by SKU/store. Quality gate: commissioned asset/location mapping; reference-weight/package-count calibration, cycle-count reconciliation and drift checks; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: COSHH/chemical inventory owner and cleaning supervisor. Precedence: physical label/container inspection and current inventory record over tag data. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation + comparison + diagnosis.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: verified container/product record and current inspection state.. Method: date/label-status rules; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: mismatch or unreadable identity becomes 'do not rely on digital identification'.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not identify unknown chemicals, declare a container safe, advise disposal/handling, or override COSHH controls. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps staff avoid taking a product whose identity or currency is already known to be unresolved.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-058 What PPE is currently listed for this approved cleaning task, and which controlled instruction says so?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target ≤ 60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: employer safety/COSHH owner and cleaning supervisor; product SDS may support. Precedence: current risk assessment/COSHH/task method and competent instruction. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: verified task, approved product and current controlled document.. Method: exact retrieval, not model-generated safety advice.. Uncertainty: conflicting/expired documents trigger supervisor/safety escalation.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not prescribe PPE beyond the authorised source, certify fit or suitability, replace training/risk assessment, or infer what a worker is wearing. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Gives staff fast access to the current authorised requirement without turning the assistant into a safety decision maker.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

6. EQUIPMENT, CHEMICALS, STORAGE AND REPLENISHMENT

CT-059 What supplies need to be replenished from central stores before the next shift starts?

Sensors and telemetry	Proposed: load-cell/smart-bin/smart-cabinet count for exact approved SKU at an authorised store; no staff tracking. Read 15-60 min and on issue/return where supported; aggregate by SKU/store. Quality gate: commissioned asset/location mapping; reference-weight/package-count calibration, cycle-count reconciliation and drift checks; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: local/central stores for quantities, cleaning supervisor for next-shift need, procurement for inbound stock. Precedence: verified count and approved minimum levels over sensor estimates. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + estimate + forecast + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: current local stock plus expected next-shift task demand.. Method: reorder-to-minimum calculation within approved stock policy.. Uncertainty: show ranges and inbound uncertainty.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not place orders without authorised workflow, substitute products, infer staff consumption or expose restricted stock details unnecessarily. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Supports a better handover by ensuring the next shift is not surprised by an avoidable stock shortage.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-060 Has any shared cleaning machine or trolley moved to another authorised store or zone since the last shift?

Sensors and telemetry	Proposed: asset-only BLE/UWB/RFID coarse zone presence for tagged machines/trolleys at authorised stores/service zones; never associate tag history with a worker. Update every 1-5 min or doorway transition; retain only operationally necessary recent location. Quality gate: commissioned asset/location mapping; tag mapping, reader health/battery and coarse-zone accuracy validation; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: equipment custodian/cleaning supervisor for asset allocation and telemetry owner for tag mapping. Precedence: explicit custody/return record and physical sighting over location estimate. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation.. Interval: at shift/task start and whenever the decisive owner-controlled status changes.. Minimum: healthy tag with recent zone-level sighting or custody record.. Method: last-known authorised zone, not continuous path.. Uncertainty: stale/ambiguous signals are labelled.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not use equipment tags to track workers, reveal secure storage to unauthorised users, or accuse someone of moving/misplacing an asset. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps locate shared machines and trolleys without turning asset tracking into staff surveillance.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

7. WASTE SEPARATION, COLLECTION AND CONTAMINATION

CT-061 Which general or recycling bins are approaching full and should be included in my next collection round?

Sensors and telemetry	Proposed: ultrasonic/time-of-flight or weight fill sensor on a specific mapped general/recycling container; no camera and no inference about contents. Read every 5-15 min with latest level and 30-60 min trend; weekly aggregation only on complete service days. Quality gate: commissioned asset/location mapping; empty/full or reference-weight calibration, obstruction/compaction checks, emptying-reset validation and drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning/waste supervisor for collection task and facilities for sensor mapping. Precedence: physical observation and scheduled collection requirement over fill estimate. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: observation + estimate + comparison + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: healthy recent reading above agreed threshold or confirmed report.. Method: threshold with persistence.. Uncertainty: impossible jumps/stale data become 'check manually'.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer who used a bin, inspect waste content with cameras, or change waste handling rules based only on fill level. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Lets the team prioritise bins likely to need collection before they overflow.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-062 Which waste points have credible evidence of overflowing before the next planned round or contractor collection?

Sensors and telemetry	Proposed: ultrasonic/time-of-flight or weight fill sensor on a specific mapped general/recycling container; no camera and no inference about contents. Read every 5-15 min with latest level and 30-60 min trend; weekly aggregation only on complete service days. Quality gate: commissioned asset/location mapping; empty/full or reference-weight calibration, obstruction/compaction checks, emptying-reset validation and drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: waste/cleaning supervisor for collection plan, waste contractor for pickup schedule. Precedence: physical check and confirmed pickup changes over forecast. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: estimate + forecast + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: at least three reliable recent fill readings plus next collection time.. Method: bounded linear/trend projection capped at sensor range. Uncertainty shown; safe failure recommends inspection when trend evidence is weak. Decision interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes. Do not convert missing coverage into a zero/false value or hide conflicts to produce a cleaner ranking.. Uncertainty: show stale, missing, unmapped and conflicting evidence separately.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not guarantee a bin will or will not overflow, infer user behaviour or alter safe handling procedures. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Supports early collection or inspection of bins most likely to become a public-facing problem.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

7. WASTE SEPARATION, COLLECTION AND CONTAMINATION

CT-063 Has contamination been reported in a recycling stream that needs the current approved handling route?

Sensors and telemetry	Proposed: the time-stamped incident/service-desk report and acknowledgement workflow; arbitrary spills, glass, sharps or contamination should not be 'detected' by speculative computer vision. A fixed wetness/alarm sensor may add evidence only at its commissioned point. Event-driven reporting; no continuous person sensing. Quality gate: commissioned asset/location mapping; workflow event integrity; any fixed alarm requires its modality-specific functional test; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: waste contractor/waste manager for stream rules and cleaning supervisor for response. Precedence: current waste-separation procedure and competent hazardous-waste instruction over automated classification. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: authorised report with bin/stream location.. Method: route to the current handling instruction for that stream.. Uncertainty: unknown item remains unknown.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not identify people who disposed of items, classify unknown hazardous material, instruct unsafe sorting or override waste-stream rules. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Warns cleaners before collection so contaminated recycling can follow the correct approved route.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-064 Which waste points from the last round still do not have a confirmed completion or authorised deferral?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning/waste task system and supervisor. Precedence: verified completion/defer record over sensor inference. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: scheduled point plus explicit non-complete/unknown state.. Method: status reconciliation.. Uncertainty: missing scan remains unknown unless policy says otherwise.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer individual productivity, accuse staff of missing a bin, or mark collection complete from a changed sensor reading alone. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps the next round pick up unresolved waste points rather than assuming the previous route was complete.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

7. WASTE SEPARATION, COLLECTION AND CONTAMINATION

CT-065 When is the next authorised waste collection, and what needs to be ready before the contractor arrives?

Sensors and telemetry	Proposed: ultrasonic/time-of-flight or weight fill sensor on a specific mapped general/recycling container; no camera and no inference about contents. Read every 5-15 min with latest level and 30-60 min trend; weekly aggregation only on complete service days. Quality gate: commissioned asset/location mapping; empty/full or reference-weight calibration, obstruction/compaction checks, emptying-reset validation and drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: waste contractor for collection schedule, waste manager/cleaning supervisor for readiness requirements. Precedence: confirmed contractor update and current site instruction over planned timetable. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + calculation + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: confirmed collection window and current required container/stream status.. Method: checklist against deadline.. Uncertainty: ETA shown as window.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not direct vehicle movements, alter contractor procedures, expose security details or promise an arrival time not confirmed by the contractor. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps the team stage the correct waste streams in time without unnecessary early movement.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-066 Where is the current approved collection point for batteries, WEEE, sharps or other special waste I may encounter?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target ≤ 60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: waste/environmental compliance owner and safety/clinical/technical owner as applicable. Precedence: current special-waste procedure and competent instruction over general building maps. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: confirmed stream classification and authorised role.. Method: rules/location lookup.. Uncertainty: unknown item gets no destination recommendation.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not identify unknown hazardous waste, tell untrained staff to handle sharps/chemicals, authorise transport, or expose controlled storage locations broadly. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Directs known special waste to the correct current collection route while protecting staff from unsafe assumptions.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

7. WASTE SEPARATION, COLLECTION AND CONTAMINATION

CT-067 Has the waste store, compactor or related equipment raised an alarm that affects today's collection route?

Sensors and telemetry	Proposed: the specific cleaning machine controller/BMS state, battery SoC/voltage/current, charger/dock state or hour counter where available; no staff location. Read 1-5 min or controller events and use only the current task/recent discharge window. Quality gate: commissioned asset/location mapping; controller point validation, SoC/energy functional checks and battery-age/drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: facilities/waste equipment owner and maintenance contractor for technical faults; cleaning/waste supervisor for collection plan. Precedence: active lockout/out-of-service instruction and competent assessment over telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation + diagnosis + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: active verified alarm/fault record.. Method: block normal-use recommendation and route to owner procedure; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: ambiguous/stale alarm remains unresolved.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not instruct staff to clear a jam, reset safety systems, enter machinery, certify equipment safe or bypass lockout controls. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Prevents routine waste handling from continuing into a known equipment fault without technical clearance.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-068 What collection route will move waste with the least disruption while staying on authorised service paths?

Sensors and telemetry	Proposed: current mapped route/door closure state plus, only as secondary context, anonymous corridor-flow counts aggregated by zone; no trajectories. Restriction events are live; flow is 5-min aggregate over the last 30-60 min. Quality gate: commissioned asset/location mapping; current route-edge/door mapping verification and manual validation of any aggregate flow counter; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: building operations/security for routes, cleaning/waste supervisor for collection plan. Precedence: access/emergency restrictions and waste-handling rules over congestion estimates. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + comparison + recommendation.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: verified start/end points and allowed route graph; congestion is optional.. Method: pathfinding with penalties for high-traffic zones.. Uncertainty: demand can change quickly.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not track individuals, block egress, use prohibited lifts/routes, expose secure areas or guarantee a route will be quiet. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps move bins and trolleys with less interference to building users while respecting waste and access rules.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

7. WASTE SEPARATION, COLLECTION AND CONTAMINATION

CT-069 Which bin fill sensors are too stale, inconsistent or unreliable to use for today's route planning?

Sensors and telemetry	Proposed: sensor-health data for each mapped bin sensor - raw level, heartbeat, calibration date, emptying resets, gap flags and paired physical checks. Operational value must be ≤ 15 min old; assess health over 24 h and 7 d. Quality gate: commissioned asset/location mapping; empty/full calibration and drift check against physical observations; clock alignment target ≤ 60 s; uptime $\geq 90\%$ over the health-assessment window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: telemetry/facilities owner for sensor health and cleaning/waste supervisor for manual checks. Precedence: physical observation for collection decisions over questionable telemetry. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: immediate/ ≤ 5 min for access, release, incident, hazard or alarm decisions. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: comparison + diagnosis.. Interval: every 15-30 minutes during the active service window, with event-driven refresh for material changes.. Minimum: recent heartbeat plus plausible measurement history.. Method: freshness/diagnostic/range checks; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: failed checks suppress the automated estimate.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not treat missing telemetry as an empty bin, conceal sensor faults or use data quality as a measure of staff performance. Must not certify safety or replace the current competent procedure or professional judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Stops unreliable sensors from creating unnecessary collection trips or missed overflowing bins.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-070 What waste volumes by stream are we actually seeing this week for planning rounds and container capacity?

Sensors and telemetry	Proposed: ultrasonic/time-of-flight or weight fill sensor on a specific mapped general/recycling container; no camera and no inference about contents. Read every 5-15 min with latest level and 30-60 min trend; weekly aggregation only on complete service days. Quality gate: commissioned asset/location mapping; empty/full or reference-weight calibration, obstruction/compaction checks, emptying-reset validation and drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: waste contractor for collected weights, waste manager for stream definitions, cleaning supervisor for rounds. Precedence: certified/contractor collection weights over sensor-estimated volumes. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + estimate + comparison.. Interval: daily or weekly planning assessment, while current urgent faults remain event-driven.. Minimum: sufficient coverage across collection days and stable stream mapping.. Method: sum measured weights; separately report estimated volume/fill with uncertainty. Safe failure avoids combining incomparable or poorly calibrated data. Decision interval: daily or weekly planning assessment, while current urgent faults remain event-driven. Do not convert missing coverage into a zero/false value or hide conflicts to produce a cleaner ranking.. Uncertainty: show stale, missing, unmapped and conflicting evidence separately.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer what individuals discarded, claim regulatory waste totals from uncertified sensors, or change contracted collection arrangements automatically. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Supports evidence-based planning of collection frequency, container capacity and staffing at stream level.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

8. QUALITY CHECKS, HANDOVER, WELLBEING AND LOWER-IMPACT OPERATIONS

CT-071 Which completed areas are due a supervisor quality check before this shift can close?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor/contract specification for quality-check requirements. Precedence: current quality standard and explicit supervisor instruction over any automated flag. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + calculation + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: completed task plus rule or reported issue requiring check.. Method: deterministic checklist selection.. Uncertainty: no telemetry-based inference of cleanliness.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not certify cleaning quality automatically, rank individual cleaners, capture people in evidence without proper basis, or replace supervisor judgement. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Ensures the right spaces receive a final quality check before handover without checking everything indiscriminately.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-072 What privacy-preserving evidence can I record to show this task is complete without photographing people, screens or sensitive material?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning contract/data-governance owner for acceptable evidence and supervisor for completion. Precedence: current evidence/retention policy over convenience. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + observation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: policy-required fields only.. Method: data-minimising checklist with optional permitted asset photo.. Uncertainty: if privacy/sensitive-content risk cannot be controlled, safe failure uses non-image evidence or supervisor confirmation.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not photograph/record people, research content, screens, documents, conversations or secure information; must not claim evidence proves more than the recorded task state. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Creates traceable completion evidence while reducing privacy and confidentiality risk.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

8. QUALITY CHECKS, HANDOVER, WELLBEING AND LOWER-IMPACT OPERATIONS

CT-073 Which cleaning-related defects or service problems keep recurring in the same place and need supervisor or facilities assessment?

Sensors and telemetry	Proposed: the time-stamped incident/service-desk report and acknowledgement workflow; arbitrary spills, glass, sharps or contamination should not be 'detected' by speculative computer vision. A fixed wetness/alarm sensor may add evidence only at its commissioned point. Event-driven reporting; no continuous person sensing. Quality gate: commissioned asset/location mapping; workflow event integrity; any fixed alarm requires its modality-specific functional test; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for issue taxonomy, helpdesk/facilities for linked faults. Precedence: verified issue records and competent technical findings over correlation. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: comparison + diagnosis + recommendation.. Interval: daily or weekly planning assessment, while current urgent faults remain event-driven.. Minimum: at least three comparable events after deduplication.. Method: count/rate by location and issue type, with exposure where available; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: recurrence does not establish cause.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not blame staff or users, infer behaviour, diagnose building faults or claim that cleaning method is the cause without evidence. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps supervisors identify places where a different schedule, asset fix or local coordination may be worth considering.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-074 What unresolved cleaning, stock, access or fault issues must I pass to the next shift?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor/handover system, with facilities/security/stores as owners of their respective statuses. Precedence: current authoritative status from each owner over prior notes. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: each open item has a verified location, owner and last update.. Method: deduplicate and group by action owner/urgency. Uncertainty stays explicit; safe failure keeps ambiguous items open rather than resolving them silently. Decision interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable. Do not convert missing coverage into a zero/false value or hide conflicts to produce a cleaner ranking.. Uncertainty: show stale, missing, unmapped and conflicting evidence separately.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not include unnecessary personal data, assign blame, disclose restricted security details or promise another team's action. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Makes the next shift immediately aware of what is unfinished, blocked or waiting on another service.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

8. QUALITY CHECKS, HANDOVER, WELLBEING AND LOWER-IMPACT OPERATIONS

CT-075 Is the planned work and travel burden for this zone materially above its normal task allowance, so I should raise it early?

Sensors and telemetry	No sensing is required for the authoritative answer. Optional QR/NFC/asset or workflow events may confirm only a mapped asset/zone ID and event time where approved. Sensor coverage, sampling, calibration and drift are not applicable. If an integration is used, require verified mapping/provenance, clock alignment (target ≤ 60 s), current source health and visible gaps; missing/stale integration data never substitutes for the owner record.
Authoritative supporting data	Owners: cleaning supervisor/contract planner for task standards and shift allocation. Precedence: current approved workload/task-duration model and supervisor judgement over analytics. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + estimate + comparison + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: current zone task list plus approved standard durations.. Method: sum standard task time against available shift window, with access/travel buffer.. Uncertainty: ranges for variable tasks.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not score or infer individual productivity, health or capability, set disciplinary conclusions, or auto-reallocate staff. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Gives supervisors evidence to adjust priorities, resources or handover when the planned work exceeds a realistic standard-time window.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-076 Can a noisy or space-consuming cleaning task be moved to a quieter confirmed window without missing the required standard or deadline?

Sensors and telemetry	Proposed: anonymous room/corridor occupancy aggregates and, only if installed, a calibrated sound-level meter storing dB(A)/Leq values rather than audio. Occupancy 5-min aggregate; sound level 1-min Leq; same-day planning horizon. Quality gate: commissioned asset/location mapping; occupancy validation and sound-level-meter calibration; never store speech/audio; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for task timing, timetable/space owner for activity schedule, facilities for any noise constraints. Precedence: explicit local restriction and required service window over occupancy estimate. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: comparison + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: task location, permitted window and nearby scheduled use.. Method: avoid high-activity intervals and respect deadlines.. Uncertainty: unscheduled use may occur.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not listen to or classify conversations, infer sensitive activity, guarantee silence, or override required cleaning/access windows. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps schedule vacuuming or machine use when it is least likely to interfere with teaching, meetings or study.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

8. QUALITY CHECKS, HANDOVER, WELLBEING AND LOWER-IMPACT OPERATIONS

CT-077 Can we reduce water or energy use on this task while still following the approved cleaning method and required standard?

Sensors and telemetry	Proposed: machine water-flow/volume counter and charger/controller energy meter for the specific approved equipment; no people sensing. Record 1-5 min samples or controller totals per task/session and aggregate only across comparable task types. Quality gate: commissioned asset/location mapping; flow/energy meter calibration, controller-total reconciliation and drift checks; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning specification/COSHH owner for method constraints, equipment manufacturer/maintenance owner for permitted settings, sustainability/facilities for resource data. Precedence: required standard, approved method and safety instructions over efficiency analytics. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + comparison + recommendation.. Interval: daily or weekly planning assessment, while current urgent faults remain event-driven.. Minimum: at least two permitted configurations with reliable consumption and outcome checks.. Method: bounded comparison.. Uncertainty: no quality equivalence means no recommendation.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not lower cleaning standards, alter chemical dilution, change safety controls or machine settings outside approved instructions, or certify environmental benefit from incomplete data. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Supports lower-impact choices only where they are already permitted and demonstrably compatible with the required service outcome.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-078 Are any consumables being used much faster than expected, suggesting a dispenser fault or stock-control issue worth checking?

Sensors and telemetry	Proposed: load-cell, time-of-flight or optical-distance level sensors on specific mapped dispensers/bins; no camera-based estimation. Read every 5-15 min or threshold event, with latest level and short rate-of-change only. Quality gate: commissioned asset/location mapping; empty/full reference calibration for each asset, refill/empty reset checks and drift monitoring; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for refill records and facilities/maintenance for dispenser faults. Precedence: physical check and confirmed fault over usage anomaly. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: comparison + diagnosis + recommendation.. Interval: daily or weekly planning assessment, while current urgent faults remain event-driven.. Minimum: reliable refill/use history across comparable days and a sustained deviation.. Method: robust baseline with fault-code correlation; any diagnosis is limited to recorded-state/data-quality triage and must not claim an unverified physical root cause.. Uncertainty: high demand may be legitimate.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not infer individual usage, protected traits or misuse by a group, and must not alter dispenser settings outside authorised maintenance. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Helps distinguish genuine high demand from a leaking, sticking or misconfigured dispenser that is wasting stock.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

8. QUALITY CHECKS, HANDOVER, WELLBEING AND LOWER-IMPACT OPERATIONS

CT-079 Can we combine supply pickups and route stops to reduce unnecessary trolley journeys without making the work late or unsafe?

Sensors and telemetry	Proposed: current mapped route/door closure state plus, only as secondary context, anonymous corridor-flow counts aggregated by zone; no trajectories. Restriction events are live; flow is 5-min aggregate over the last 30-60 min. Quality gate: commissioned asset/location mapping; current route-edge/door mapping verification and manual validation of any aggregate flow counter; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for route/tasks, stores for stock, security/building operations for authorised routes. Precedence: access/restriction and task deadline over route optimisation. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: calculation + estimate + recommendation.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: known tasks, needed supplies and verified stores.. Method: group pickups by route while respecting load/access constraints.. Uncertainty: stock or release changes.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not track staff, require unsafe trolley loads, route through restricted areas or make automatic workload decisions. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Reduces avoidable walking and trolley movement while keeping the route within normal operational boundaries.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

CT-080 At the end of the shift, are all assigned tasks either confirmed complete, explicitly deferred or escalated to the right current owner?

Sensors and telemetry	Proposed: cleaning-workflow task/complete/defer/escalate events for current assigned task IDs; optional zone QR/NFC confirmation, not worker location. Coverage: current/previous shift only; event-driven timestamps, aggregated only for shift status. Quality gate: commissioned asset/location mapping; event integrity and scanner/reader functional verification; numerical calibration N/A; clock alignment target ≤ 60 s; uptime $\geq 80\%$ of decision window; show gaps, drift/outliers and device/gateway/processing provenance. Failed gates mean unavailable/unknown; never zero or last-known-as-current.
Authoritative supporting data	Owners: cleaning supervisor for shift closure, with facilities/security/stores/waste owners for delegated issues. Precedence: explicit authoritative status from each owner over inference. Times: keep effective/valid, observed/entered and retrieved distinct; show role permission and source/version provenance. Freshness: current shift/day for operational routing; complete, explicitly dated periods for weekly trend/planning. Conflict: current responsible-owner records and competent direct observations within scope outrank telemetry; active safety/security uses the more restrictive valid instruction until resolved. May 2021 plan: historical orientation only; never current authority.
Analysis required	Type: lookup + calculation + comparison.. Interval: on the relevant task, inspection, handover or status event, with a shift-end reconciliation where applicable.. Minimum: every planned task has an explicit valid state and unresolved item has an owner/action route.. Method: completeness check and exception list.. Uncertainty: unknown remains open.. Safe failure: if minimum evidence, permission or quality is unmet, state what cannot be determined, exclude the weak input, and give the current owner or physical-check escalation rather than guess.
Answer boundary	Must not silently close tasks, infer why work remains, certify building cleanliness/safety, or promise that another team will resolve escalated items. Across all answers: no identification/tracking, behaviour/productivity/protected-trait inference, conversation or sensitive-content monitoring, access granting, chemical/PPE prescription, safety/cleanliness certification, equipment control, incident closure, or replacement of current cleaning standards, COSHH/SDS, security rules, competent professionals or supervisor judgement.
Why this is useful	Provides a defensible shift close and a precise handover instead of leaving ambiguous work hidden in separate systems.

VERIFY CURRENT AUTHORITY AND EVIDENCE BEFORE USE

Catalogue summary

This catalogue supports operational choices that cleaning and caretaking teams genuinely make. Mandatory standards and current authority stay ahead of sensing; telemetry is used only where its physical modality can support the claim; uncertainty and safe failure remain explicit.

No.	Operational category	Questions	Primary decisions supported
1	shifts, zones, standards and daily priorities	CT-001 - CT-010	shift sequencing, standards, due work and handover
2	occupancy-driven demand and room-release information	CT-011 - CT-020	release checks, cleanable windows and demand-led physical checks
3	washrooms, consumables and amenities	CT-021 - CT-030	washroom rounds, consumables, amenity faults and route stock
4	spills, damage, faults and urgent escalation	CT-031 - CT-040	incident facts, escalation, facilities faults and wet-weather checks
5	keys, access, lone working and security boundaries	CT-041 - CT-050	access prerequisites, authorised routes, lone-working controls and custody
6	equipment, chemicals, storage and replenishment	CT-051 - CT-060	serviceable equipment, controlled methods, stores and replenishment
7	waste separation, collection and contamination	CT-061 - CT-070	bin rounds, overflow risk, special waste, collection routes and volumes
8	quality checks, handover, wellbeing and lower-impact operations	CT-071 - CT-080	QA, handover, workload escalation, lower-impact operation and route consolidation

SAFE DEFAULT

When a decisive owner record is missing, stale, conflicting, permission-limited or unmapped, state what cannot be determined, exclude the weak input from ranking/forecasting, and give the current owner or physical-check escalation. Never fill the gap by inferring people, permission, safety, chemical choice or task completion.

Evidence and permissions rules

1 Verify deployment before use

Treat every sensor, gateway, integration and analytic as proposed until deployment, mapping, commissioning, permission and current health are independently verified.

2 State what sensing can physically observe

Name modality/location, footprint, sampling/aggregation and blind spots. A point wetness sensor is not a whole-floor inspection; a fill sensor does not classify waste; an occupancy count is not room release.

3 Carry a telemetry quality gate

Report calibration/functional verification where applicable, clock alignment, uptime, gaps, drift/outliers and device-gateway-processing provenance. Failed gates produce unavailable/unknown, never zero.

4 Preserve time semantics and provenance

For owner records, distinguish effective/valid, observed/entered and retrieved times. Keep role permission and source/version provenance visible. Label staleness rather than silently reusing old data.

5 Resolve conflicts by authority

Do not average contradictions. Current responsible-owner records and competent direct observations within scope outrank weaker telemetry. Active safety/security uses the more restrictive valid instruction until resolved.

6 Make analysis traceable

Name operation type, decision interval, minimum evidence, method and uncertainty. Forecasts need adequate complete windows; comparisons need like-for-like baselines; diagnosis must not claim unsupported cause.

7 Fail safely and honestly

If decisive evidence, permission or current mapping is missing, say "cannot determine" and identify the authorised owner/physical check. Do not substitute historical plans, stale telemetry or convenient inference.

8 Keep people and controlled procedures outside inference

No identification/tracking, behaviour/protected-trait inference, conversation monitoring, access granting, chemical/PPE prescription, hazard classification, cleanliness/safety certification, equipment control, or replacement of standards, COSHH/SDS, security, emergency or supervisor judgement.

HISTORICAL SPATIAL REFERENCE

The May 2021 floor plan can support historical orientation only. Current room function, service mapping, routes, stores, access, hazards, closures and operational responsibility require a current authoritative owner.